National University of Science and Technology
POLITEHNICA Bucharest
Faculty of Engineering in Foreign Languages

Data and Signal Processing for Business – Project

Agent-Based SIR Simulation of Rumor Spreading
in Social Networks

Jaime MORALES CASTRO

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1 Introduction

The goal of this project is to simulate how a rumor spreads through a social network using an Agent-Based Model. The model is based on the SIR framework, which was covered in the course lectures as a way to describe how something propagates through a population. Instead of a disease, in this case the thing spreading is a piece of information.

The reason for choosing this topic is the economic consequences that misinformation can have. When false news circulates in a network, it can affect financial markets, change how consumers behave, and damage company reputation. This makes rumor diffusion relevant as an economic model, not just as a communication problem.

To run the simulation, we use Python with the MESA library, which is a framework designed for Agent-Based Models. Two different network topologies are compared: Barabási-Albert and Watts-Strogatz. The results show how the structure of the network affects how fast the rumor spreads and how many people it reaches.

2 Background

2.1 The SIR Model

The SIR model is a classical model that describes how something spreads through a population. It divides everyone into three groups: Susceptible (S), Infected (I), and Recovered (R). People move from S to I when they come into contact with someone infected, and from I to R when they recover. Two parameters control the behavior: β is the transmission rate and γ is the recovery rate.

The model was introduced in Lecture 7 of the course. In the original formulation, the dynamics are described by a set of ordinary differential equations:

$$\frac{dS}{dt} = -\beta \cdot \frac{S \cdot I}{N} \tag{1}$$

$$\frac{dI}{dt} = \beta \cdot \frac{S \cdot I}{N} - \gamma \cdot I \tag{2}$$

$$\frac{dR}{dt} = \gamma \cdot I \tag{3}$$

where N is the total size of the population. These equations assume that any individual can interact with any other individual at any time, which is a simplification that does not really work for social networks. This is the main reason for using an Agent-Based approach instead, because it allows to place agents on a network and only let them interact with their direct neighbors.

2.2 Rumor Spreading and Economic Impact

Adapting the SIR model to rumors is quite natural. A Susceptible agent has not heard the rumor yet. An Infected agent has heard it and is actively sharing it with others. A Recovered agent heard it at some point but lost interest and stopped spreading it.

The connection to economics is clear. Vosoughi et al. [1] studied how news spreads on Twitter and estimated that the base probability of someone sharing information after receiving it is around 5 to 15 percent. False or novel content tends to spread faster than

true news, which means that in a network like a financial platform, a false rumor about a company can reach most users before any official correction is made. The consequences for stock prices or consumer behavior can be significant.

3 Model Description

3.1 Network Topologies

A key part of this simulation is comparing two different network structures to see how they affect spreading dynamics.

The Barabási-Albert (BA) network is a scale-free network. This means a small number of nodes have a lot of connections (hubs), while most nodes have very few. This is similar to how real social networks like Twitter or Instagram work, where a few users have millions of followers and the rest have much smaller audiences. The network is generated with $m = 2$, meaning each new node connects to 2 existing nodes when the network is built.

The Watts-Strogatz (WS) network is more homogeneous. Every node has roughly the same number of connections, and there are no extreme hubs. It is generated with $k = 4$ nearest neighbors and a rewiring probability of $p = 0.1$. This structure is more similar to a closed professional community, where people tend to know a similar number of contacts.

3.2 Agent Behavior

Each agent has a state that can be S, I, or R. At each time step, agents in state I do two things. First, they go through all their neighbors in the network. For each neighbor in state S, they draw a random number and compare it to β. If the draw is below β, that neighbor becomes infected. Second, the agent draws against γ to decide if it recovers and moves to state R.

Agents in state S and R do not take any action. The simulation starts with all agents in state S except one, which is randomly chosen to be the first spreader.

4 Implementation

The simulation is built in Python using the MESA library. NetworkX is used to generate the network topologies. The code has two main classes.

The `RumorAgent` class represents each person in the simulation. It has a `state` attribute (S, I, or R) and a `step()` method that contains the spreading and recovery logic.

The `RumorModel` class handles the setup. It creates the network, places one agent on each node, and sets one randomly chosen agent to state I at the start. A `DataCollector` object from MESA records how many agents are in each state after every step.

The parameters used are: $N = 100$ agents, $\beta = 0.10$, $\gamma = 0.05$, and a random seed of 42 to make results reproducible. The value $\beta = 0.10$ is taken from Vosoughi et al. [1], who report a base spreading probability of approximately 5 to 15 percent per exposure. Each time step represents one day, and the simulation runs for a maximum of 80 days.

5 Results

Figure 1 shows the SIR dynamics in the Barabási-Albert network. The susceptible curve drops quickly, and the infected curve reaches a peak of 53 agents around day 26. By the end, 90 out of 100 agents have heard the rumor. The fast rise of the infected curve is caused by hub nodes: once the rumor reaches one of them, it spreads to a large number of agents very quickly.

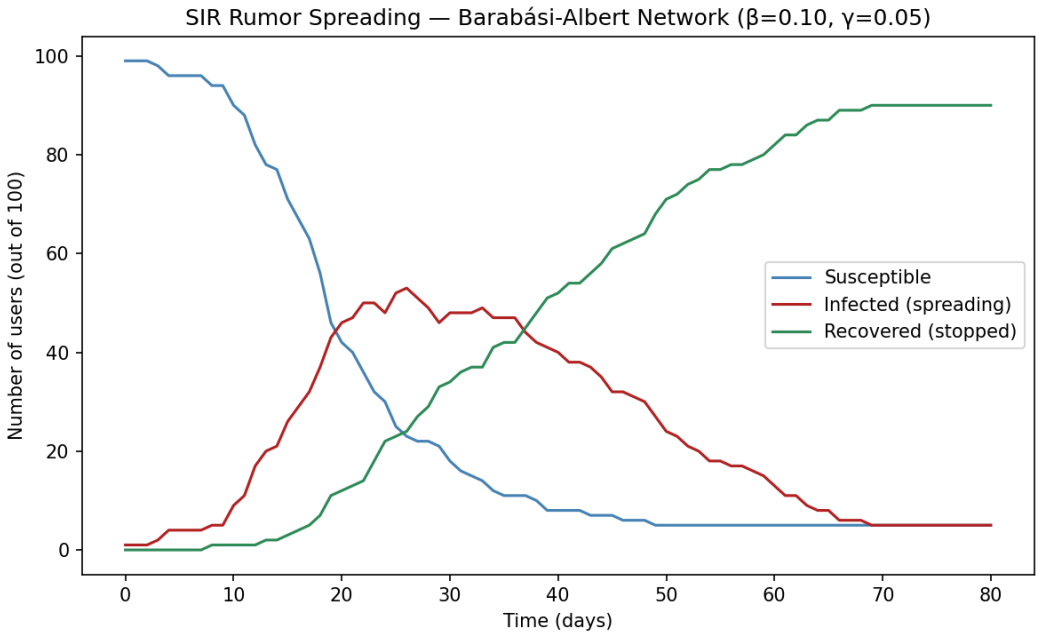


Figure 1: SIR dynamics in a Barabási-Albert network ($\beta = 0.10$, $\gamma = 0.05$, $N = 100$).

Figure 2 shows the same simulation with a Watts-Strogatz network. The dynamics are noticeably slower. The infected curve never goes above 36 agents and peaks around day 54. The final reach is also lower, at 80 out of 100 agents. Without hub nodes, the rumor can only spread step by step through the network with no sudden jumps.

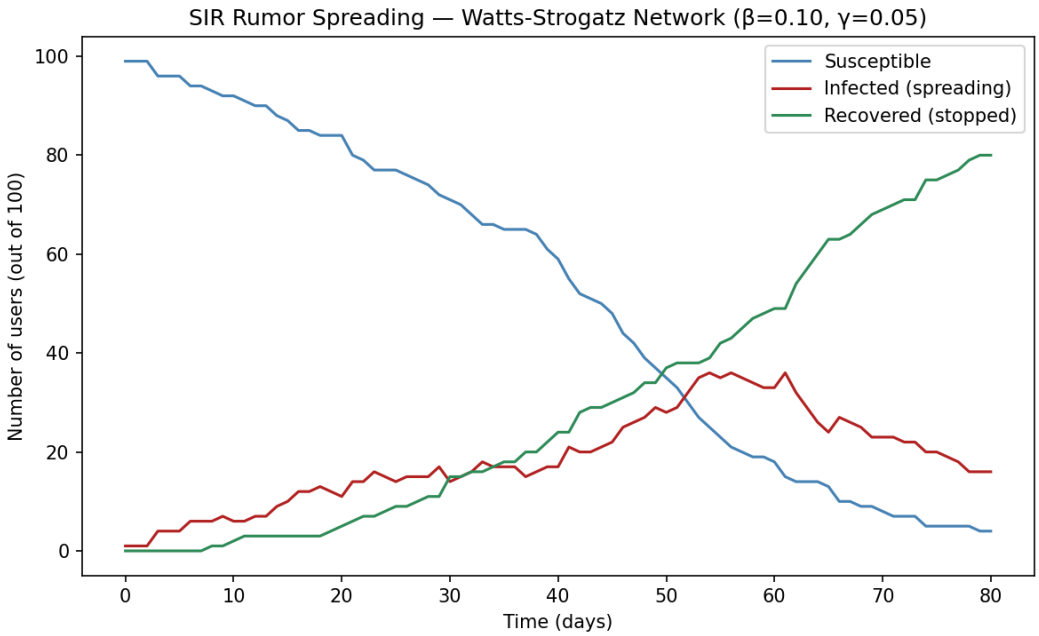


Figure 2: SIR dynamics in a Watts-Strogatz network ($\beta = 0.10$, $\gamma = 0.05$, $N = 100$).

Figure 3 shows both infected curves together. It is easy to see that the BA network produces a higher and faster peak. Another interesting thing is that when the BA curve is already declining, the WS curve is still rising. The two networks not only differ in how high the peak is, but also in when it happens.

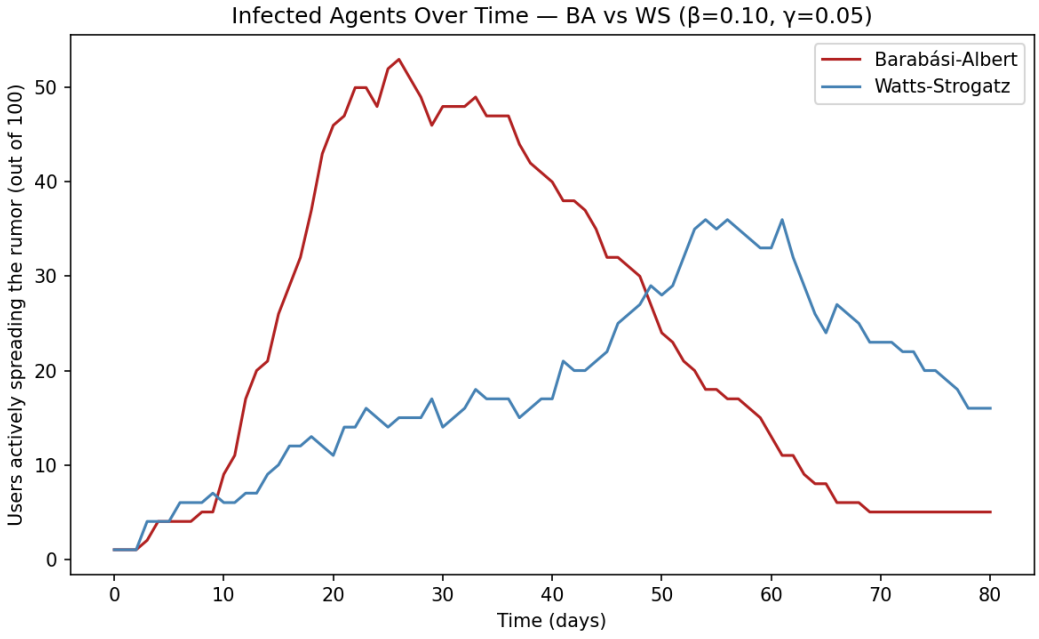


Figure 3: Comparison of infected agents over time in BA and WS networks.

Figure 4 summarizes the three main metrics for each network. In all categories, the BA network reaches higher values and does so faster than the WS network.

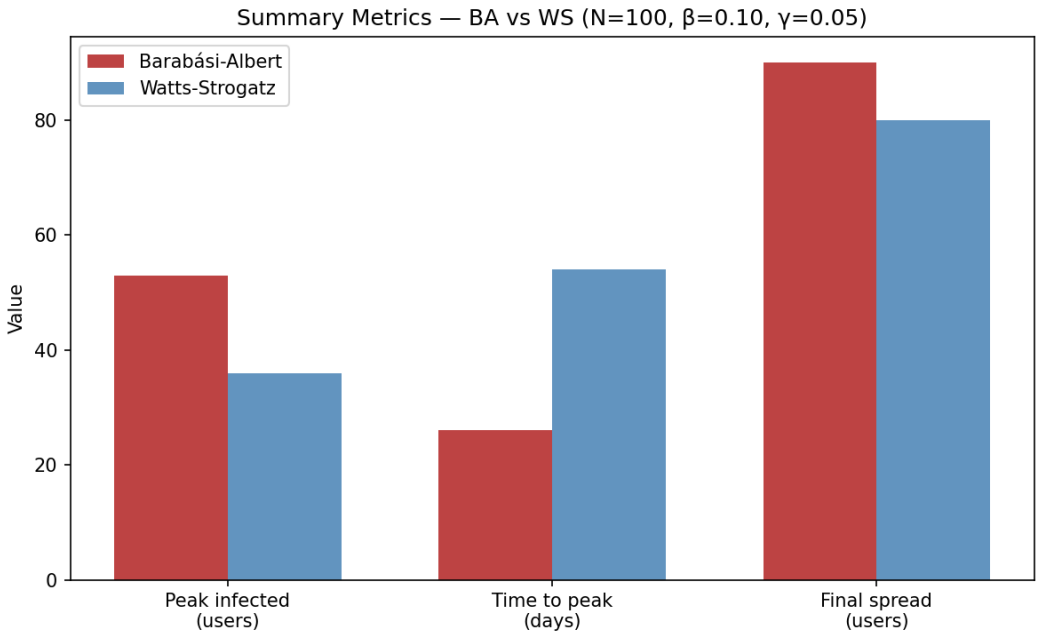


Figure 4: Summary metrics comparing BA and WS networks ($N = 100$, $\beta = 0.10$, $\gamma = 0.05$).

6 Discussion

The results show that network topology has a big effect on how a rumor spreads. The BA network, because of its hub nodes, allows the rumor to reach most of the population much faster than the WS network.

From an economic point of view, this has practical consequences. A platform like Twitter or a financial news site, which is closer to a BA structure, is much more vulnerable to misinformation. If a false rumor about a company reaches an influencer node early, it can spread to most of the network in just a few weeks, before any correction can be published. In the simulation, the BA network peaks on day 26, while WS peaks on day 54. That is almost a month of difference, which in a real scenario involving financial markets would be very significant.

A WS-type network, like a closed professional community, gives more time to react. The slower spreading and lower final reach (80% vs 90%) mean that fact-checks or official statements have a better chance of reaching people before the rumor does.

The main limitation of this model is that all agents share the same value of β. In reality, some people are more likely to share information than others, and hub nodes effectively reach more people just because they have more connections. A more realistic version of the model could use different β values for different agents.

7 Conclusion

This project showed how the SIR model can be adapted to study rumor spreading in social networks using an Agent-Based approach. The simulation was run on two different network types and the results were compared.

The main conclusion is that network structure matters a lot. In the Barabási-Albert network, the rumor spreads faster and reaches more people than in the Watts-Strogatz network. This is because the hub nodes in BA act as accelerators when the rumor reaches them.

From an economic perspective, this means that platforms with a few very connected users are at higher risk from misinformation. The short time window before the rumor peaks in BA networks makes it much harder to intervene in time.

The model has simplifications that could be improved in future work, mainly the assumption that all agents behave the same way. Adding heterogeneous spreading probabilities or testing more network types would make the results more realistic.

References

[1] Vosoughi, S., Roy, D., & Aral, S. (2018). The spread of true and false news online. *Science*, 359(6380), 1146–1151. Available at: https://mitsloan.mit.edu/ideas-made-to-matter/study-false-news-spreads-faster-truth